

# CONTROL SYSTEMS LABORATORY

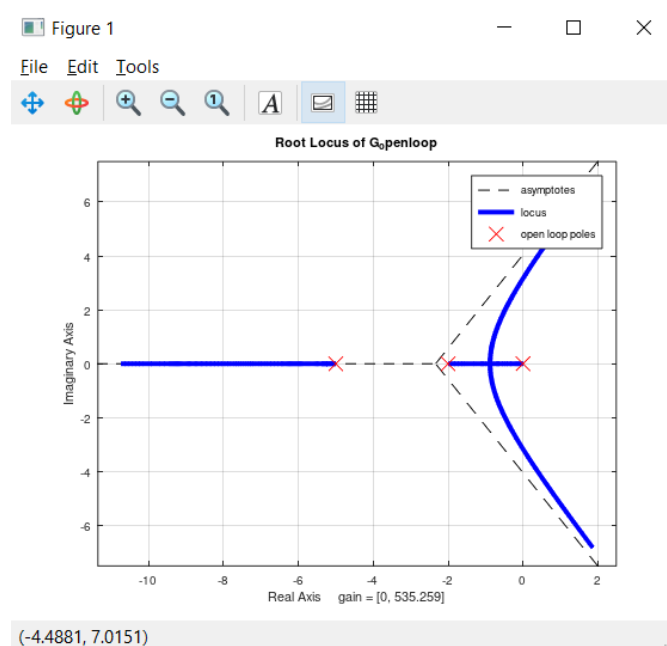
|                         |               |
|-------------------------|---------------|
| <b>Experiment No:</b>   | 5             |
| <b>Experiment Name:</b> | EXPERIMENT 5  |
| <b>Student No:</b>      | 151220192027  |
| <b>Student Name:</b>    | Mustafa Yağcı |
| <b>Date:</b>            | 25.05.2023    |

## 1. OBJECTIVES

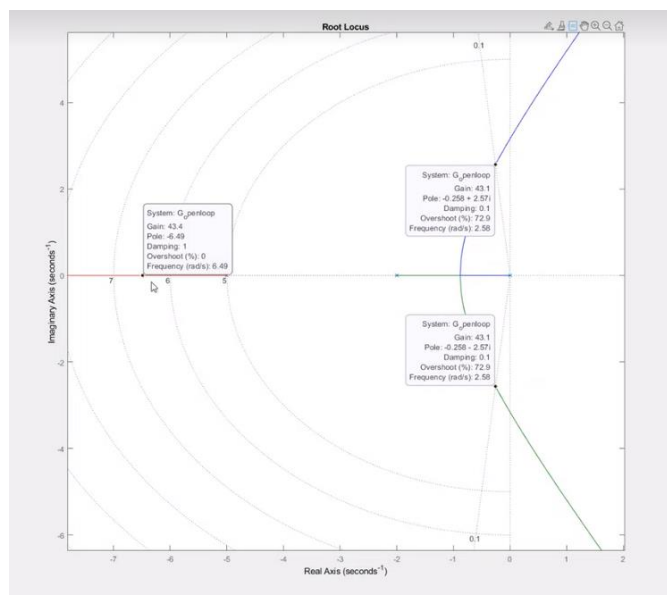
1. Obtain transient response analysis by using Octave.
2. Obtain steady-state response analysis by using Octave.
3. Explain the controller design strategies for a given dynamics.

## 2. EXPERIMENTAL RESULTS

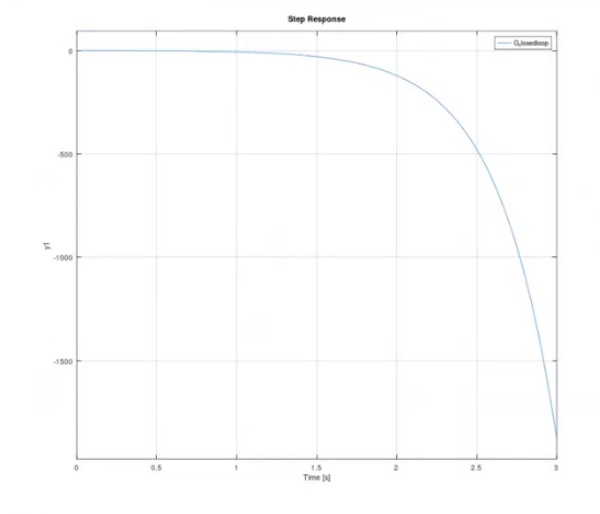
1- Rlocus =



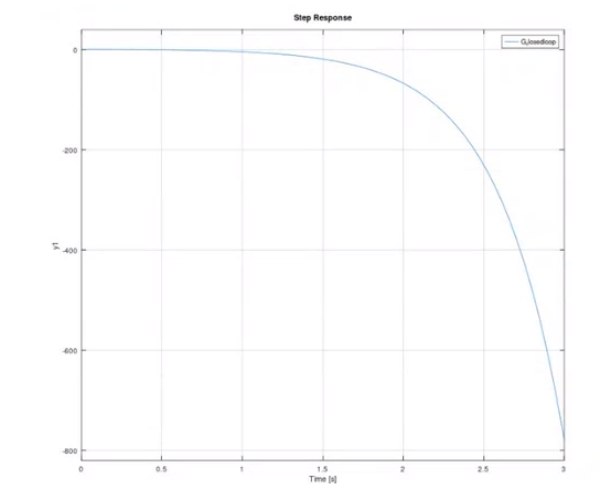
2-



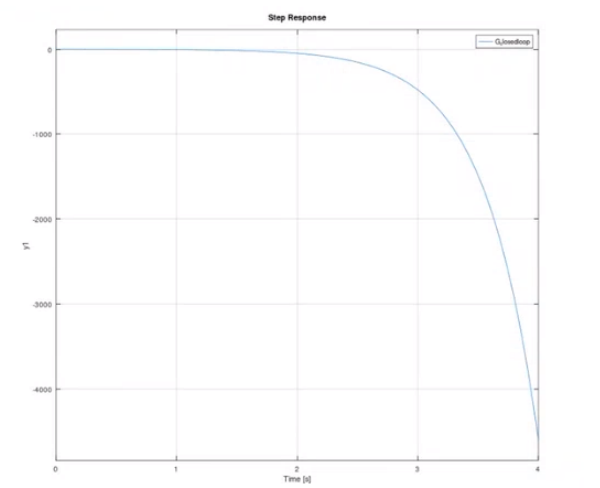
3-  $K=-100$



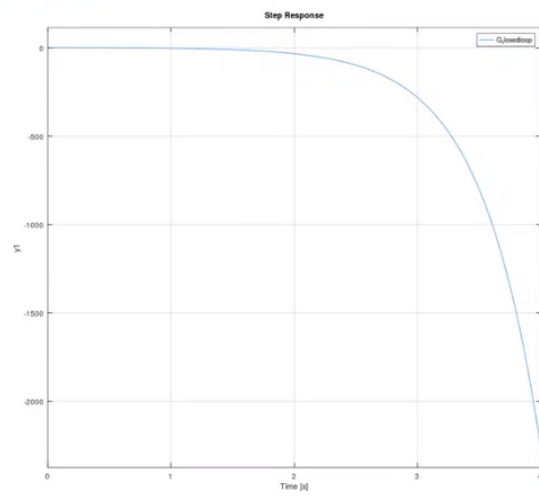
$K = -90$



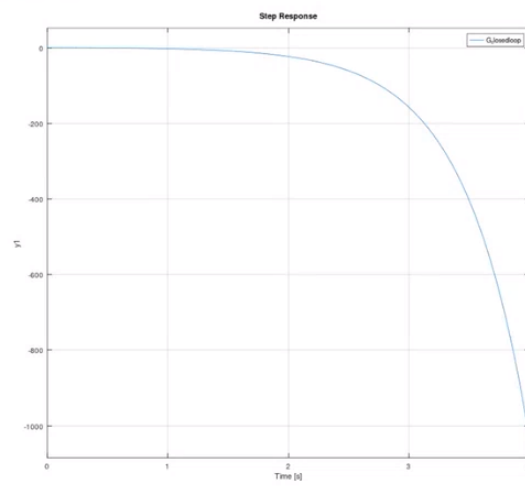
$K = -80$



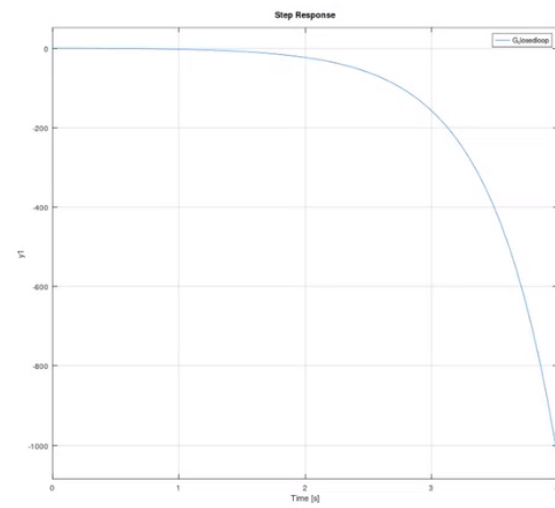
$K=-70$



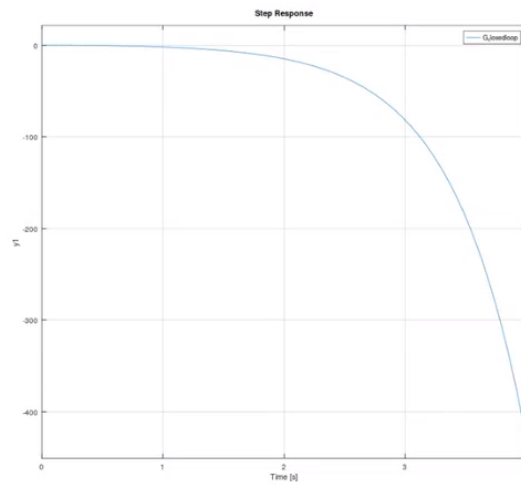
$K=-60$



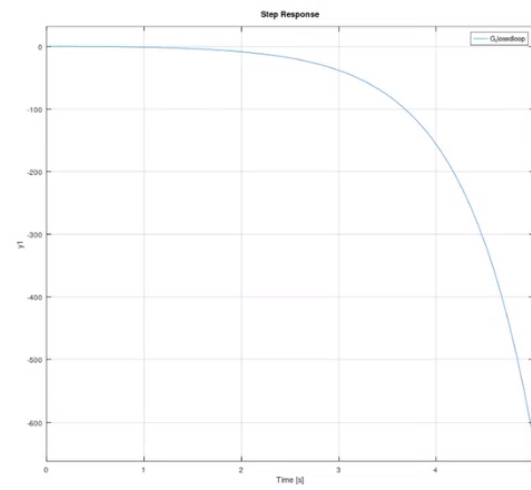
$K=-50$



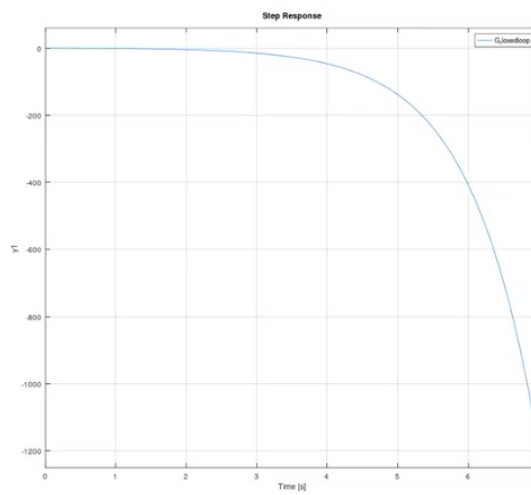
$K=-40$



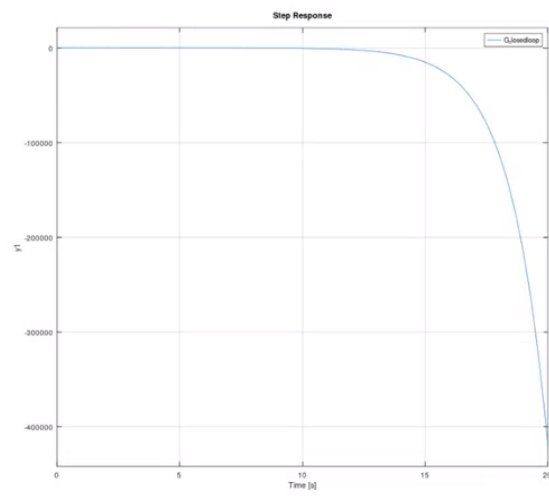
$K=-30$



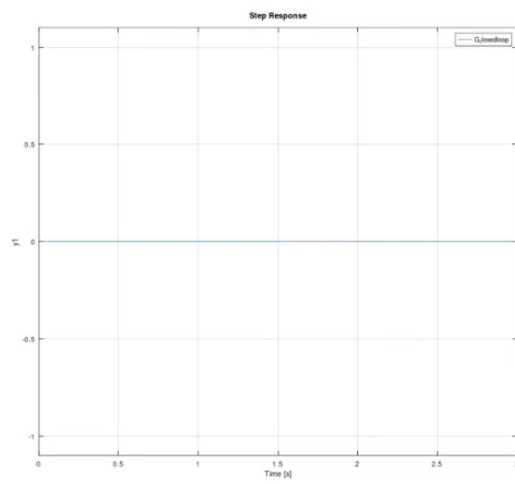
$K=-20$



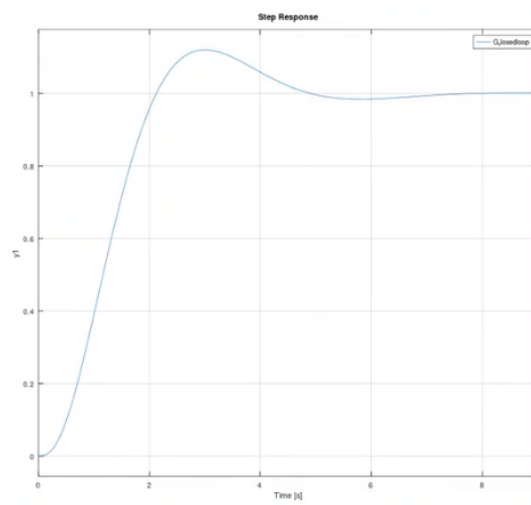
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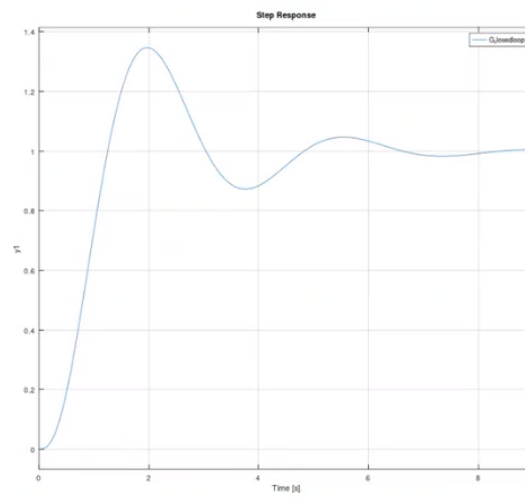
$K=0$



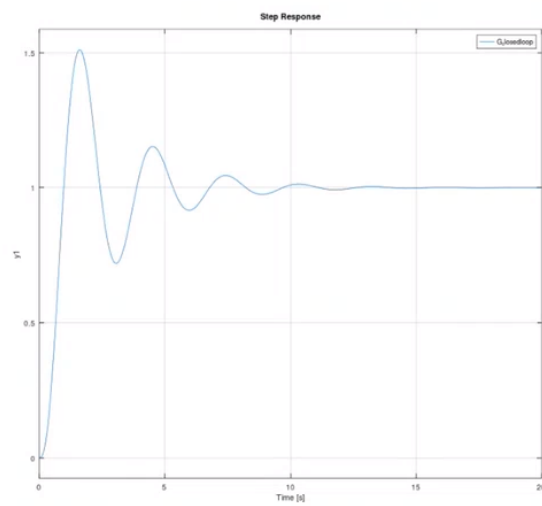
$K=10$



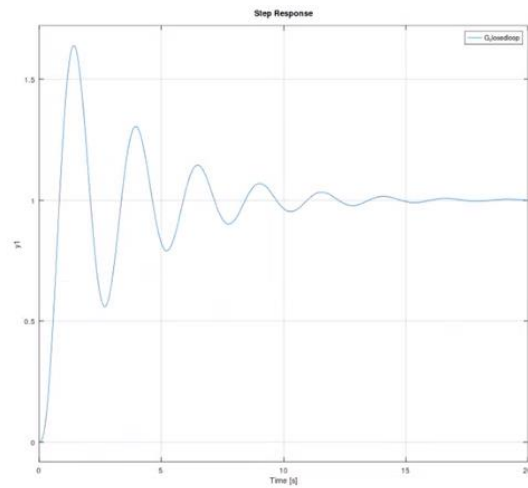
K=20



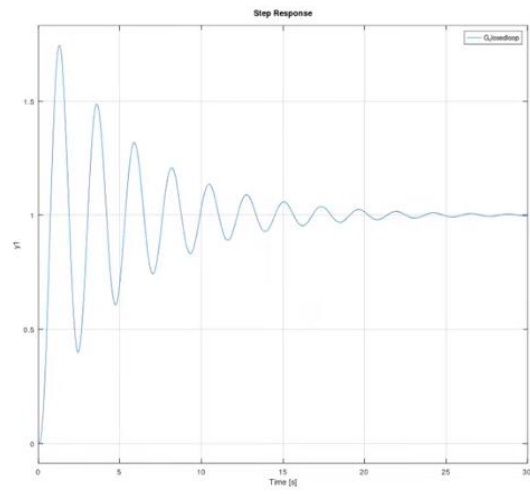
K=30



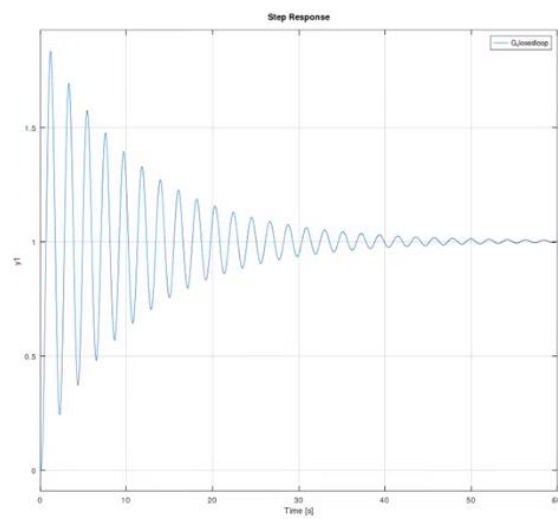
K=40



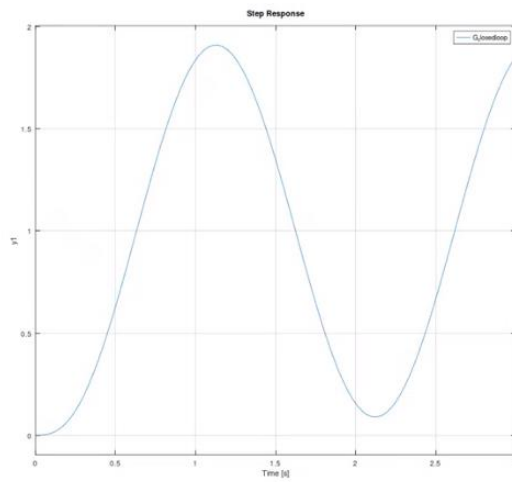
K=50



K=60

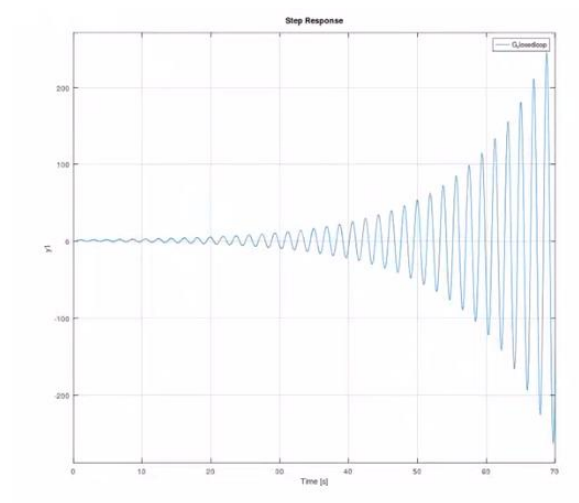


K=70

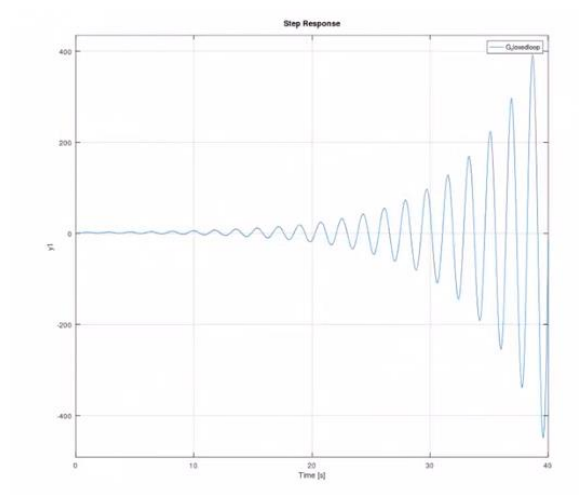




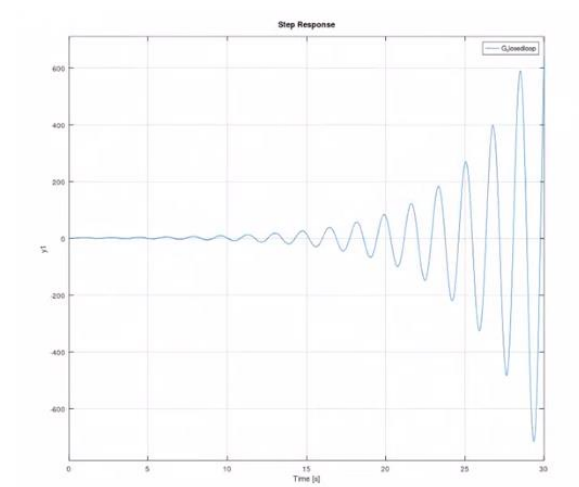
K=80



K=90



K=100



If the desired system is on the root locus, we can find the corresponding value of  $k$  that achieves this and use a P-type controller.

### **3. CONCLUSION & DISCUSSION**

In conclusion, the system is unstable for  $k$  values ranging from -100 to 0. After -50, the root continues to decrease, and the system remains unstable. In the range of  $k$  values from 10 to 60, the oscillation increases, but the system reaches a steady state without becoming unstable. When the  $k$  value is 70, the system has a stable margin. For  $k$  values above 70, the system becomes unstable.

### **REFERENCES**

Write your reference list in this section by using numbering with square brackets - []. You should follow IEEE Citation StyleGuide for referencing format.

- [1] IEEE Citation Style Guide, Available: <http://www.ijssst.info/info/IEEE-Citation-StyleGuide.pdf>, Access Date: October 2021